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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Triple T
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG-8 Decent Work and Economic Growth
Project / Solution Name	GreenReady
Name (Leader)	Remy Ong Jing Yi
Email & Phone (Leader)	remyongjingyi1233@gmail.com / +60126406646
University Affiliation(s)	Asia Pacific University of Technology & Innovation
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia has 1.08 million micro and small enterprises (MSMEs produced 39.7% of GDP and 8.09 million jobs in 2025). Every month each one receives a TNB bill, but owners lack the time and expertise to check whether the charges are normal or which upgrade would cut the cost. Electricity costs are rising: energy-intensive SMEs reported up to 10% higher monthly bills in 2025, and up to RM9,000 more a month in industrial zones.

Three problems follow. (1) Checking the bill by hand is slow, so most owners skip it and notice a spike only when cash runs short. (2) Upgrade decisions are made blind: quotes arrive in different formats from sellers who push their own product, so owners cannot tell whether LED lighting, inverter aircon or solar pays back first. (3) There is an evidence gap: a professional carbon audit costs RM20,000–50,000, so SMEs cannot prove their savings, green financing stays out of reach, and banks and suppliers struggle to find ready SMEs.

Our serviceable market is 195,769 manufacturing and food & beverage firms, the most electricity-intensive groups. The issue matters economically (rising costs for small firms), environmentally (avoidable kWh and CO_2e at Malaysia's grid factor of 0.740 kg CO_2e/kWh) and socially: small firms are shut out of the green transition by cost and knowledge, not by lack of intent.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

GreenReady is a module inside the Orbyte SME hub that turns the monthly TNB bill into decisions and connections. The owner forwards a bill on WhatsApp or uploads a photo, PDF or screenshot.

Where AI is applied: a vision-language model (VLM) reads every charge line and gives each field a confidence score; fields below 85% go to staff for correction. AI only reads documents. Fixed rules do every check and calculation, so the same inputs always give the same answer.

What the system does: (1) 8 fixed bill checks against the official tariff, a spike alert and a 6-month trend; (2) a fair comparison of LED, inverter aircon and solar on the same formulas (cost, yearly

saving, payback, CO2e); (3) a Preparation Pack PDF of six months of checked bills; (4) with the owner's approval, an anonymous listing where banks see a criteria match and suppliers see leads in their category and area. Contact details unlock only after the owner accepts.

Orchestration: a LangGraph workflow runs the journey with two owner-approval gates, and every upload, approval and request is written to an activity log.

Workshop skills used: GenAI APIs (VLM document extraction on RunPod), database architecture (Supabase for data, storage, roles and the activity log) and cloud deployment (FastAPI backend hosted on Replit).

Boundaries: no credit scores, ratings or loan advice, and no money moves on the platform.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Stack.

Intake: WhatsApp Business Platform and web upload (photo, PDF, screenshot).

AI extraction: open vision-language model served on RunPod serverless GPU; returns each charge line with a confidence score.

Backend and orchestration: Python FastAPI hosted on Replit; LangGraph workflow with human-approval gates.

Rules engine: deterministic Python module for the 8 bill checks, spike alert, and savings, payback and CO2e formulas (TNB tariff, grid factor 0.740 kg CO2e/kWh).

Database and security: Supabase (PostgreSQL, file storage, login and roles, activity log); anonymised listings.

Interface: web dashboards for SME owner, investor and supplier; Preparation Pack generated as PDF.

Block diagram (flow).

TNB bill (WhatsApp / upload) → Supabase storage → FastAPI on Replit → RunPod VLM extraction (confidence score) → low-confidence fields to staff review → rules engine (8 checks, spike alert, upgrade comparison) → Preparation Pack → Owner approval 1 → anonymous listing on Opportunity Board → investor or supplier request → Owner approval 2 → details unlock, EOI or outreach draft. LangGraph coordinates every step and each step is logged in Supabase.

Sponsor-provided GenAI API or cloud platform: Replit, used for development and hosting the website. Runpod is used as a serverless GPU for VLM

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we mapped five user groups (SME owner, SME staff, bank, supplier, Orbyte) and gathered published evidence on their pain points from SAMENTA, UOB and Malaysian business press.

Define: three problems: the recurring bill-check burden, blind upgrade decisions, and the evidence gap that blocks funding.

Ideate: we compared Trexon, buySolar, UOB U-Energy and myTNB, found that none offers a seller-free comparison of all upgrades with consent-based matching, and chose that as our focus.

Prototype: a deliberately narrow scope: Peninsular LV tariffs, three upgrades and synthetic company, bank and supplier data. AI reads documents, rules calculate, and humans approve. Our sample bill already passes 8 of 8 checks.

Test and deploy: before Phase 2 we will measure time per bill (manual vs GreenReady) and extraction accuracy on 10 test bills. Deployment uses managed services (Replit, Supabase, RunPod), works through WhatsApp so there is no new app to learn, and has a staff-review step and manual-entry fallback for misread bills.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 8.2 (main):** higher economic productivity through technological upgrading and innovation. A manual monthly bill review becomes a forward-and-read task. **Target 8.3 (main):** support the growth of micro, small and medium enterprises, including access to financial services. SMEs can plan upgrades and reach green funders and suppliers without a consultant. **Co-benefits:** Target 9.3 (small enterprises' access to financial services) and Target 7.3 (energy efficiency); also supports SDG 13.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Energy-intensive SMEs report up to 10% higher monthly electricity bills, and up to RM9,000 more a month in industrial zones. A professional carbon audit costs RM20,000–50,000], and owners spend about 11 hours a week on admin and finance tasks (UK survey, used as a proxy). Serviceable market: 195,769 firms.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Minutes to check one TNB bill, manual vs GreenReady. Field-level extraction accuracy on 10 test bills. Bill checks passed (out of 8). kWh, RM and kg CO2e saved per modelled upgrade, using the 0.740 kg CO2e/kWh grid factor .
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	At least 80% reduction in time to check one bill; at least 95% extraction accuracy on 10 test bills (fields under 85% confidence routed to staff); 8 of 8 bill checks passed on every

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	test bill. Modelled impact of one typical upgrade: LED RM1,366 and 2.0

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name & Version	Purpose in Project	License
Python	Backend application and data-processing logic	Python Software Foundation License
FastAPI	HTTP API and webhook server	MIT
LangGraph	Workflow orchestration and human-approval checkpoint	MIT
Pydantic	Validation of AI-extracted document fields	MIT
Supabase Python Client	Database and object-storage access	MIT
Google API Python Client	Gmail API integration	Apache-2.0
Google Auth / Google Auth OAuthlib	Gmail OAuth authentication	Apache-2.0
PyMuPDF	PDF and document conversion before AI processing	AGPL-3.0 or commercial license

Name & Version	Purpose in Project	License
Requests	HTTP communication with external services	Apache-2.0
python-dotenv	Loading environment variables locally	BSD-3-Clause
HTML5, CSS3, JavaScript	Planned web frontend	Standards / project-authored code

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24th October 2026